From Text to Space: Mapping Abstract Spatial Models in LLMs during a Grid-World Navigation Task

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Abstract. Understanding how large language models (LLMs) represent and reason about spatial information is crucial for building robust agentic systems that can navigate real and simulated environments. In this work, we investigate the influence of different text-based spatial representations on LLM performance and internal activations in a grid-world navigation task. By evaluating models of various sizes on a task that requires navigating toward a goal, we examine how the format used to encode spatial information impacts decision-making. Our experiments reveal that cartesian representations of space consistently yield higher success rates and path efficiency, with performance scaling markedly with model size. Moreover, probing LLaMA-3.1-8B revealed subsets of internal units—primarily located in intermediate layers—that robustly correlate with spatial features, such as the position of the agent in the grid or action correctness, regardless of how that information is represented, and are also activated by unrelated spatial reasoning tasks. This work advances our understanding of how LLMs process spatial information and provides valuable insights for developing more interpretable and robust agentic AI systems.

Keywords: Spatial Navigation · LLM Agents · Abstract World Models

1 Introduction

Large language models (LLMs) have demonstrated impressive capabilities in processing and generating text [1, 2], yet a critical question remains: do these models develop abstract internal representations of the world, or do they simply memorize typical reasoning paths? This debate is particularly heated in non-textual domains—such as spatial or perceptual tasks—where the models have only been exposed to such concepts indirectly through text [3–5]. While several studies have shown that LLMs can learn and solve tasks in these domains [6–8], and that their internal representations can be linearly mapped to those external structures [9], a clear understanding of how these internal models are composed and how they influence behavior is still lacking.

This question is of increasing importance as LLMs are being deployed as agents that interact with environments by taking sequential actions based on textual inputs [10, 11]. In non-multimodal settings, the representation of spatial information as text is crucial, as language models are known to be highly sensitive to prompting [12–14]; the format in which this information is encoded may profoundly influence a model's ability to extract relevant knowledge and make correct decisions. Furthermore, studying distinct representations of spatial information can help unveil if and how LLMs encode abstract models of space (i.e. internal activations that contain spatial information and are invariant to changes in prompting and context), and whether they leverage those internal world models to make decisions.

To address these issues, we investigated how LLM behavior and understanding of space change based on how spatial information is provided in the prompt, and whether there exists an internal model of space that is invariant to how that information is represented. We evaluated the LLaMA-3 family of models [15], spanning model sizes from 1B to 90B parameters, on a Grid-World Spatial Orientation Task (GWSOT). In this task, models were required to navigate a 2D grid toward a goal by selecting one of four possible moves. We tested several different Spatial Information Representations (SIRs) in text form. We categorized these representations into three types: Cartesian representations, in which the (x, y) coordinates of both the agent and the goal are explicitly encoded; Topographic representations, which preserve the grid-like spatial structure in the text; and Textual representations, which describe the world state in prose. Across model sizes, we found that Cartesian representations consistently yielded better performance.

Furthermore, to gain insight into the internal processing of spatial information, we probed the activations of the LLaMA-3.1-8B model during the GWSOT. Our analysis revealed parameters that significantly predicted the position of the agent in the grid, as well as parameters that predicted action correctness in the subsequent step, regardless of how spatial information was represented in the prompt. Intriguingly, a subset of these units, primarily located in the model's middle layers, were also more active when the model tackled spatial reasoning questions in unrelated contexts. These findings point to the existence of core units that form an internal spatial model invariant to prompt variations, context, and task specificity.

Our contributions can be summarized as follows:

- We demonstrate that spatial orientation performance scales with model size.
- We show that specific ways of encoding spatial information have significant effects on model performance, even when the conveyed information is equivalent.
- We reveal that LLMs can represent the position of an agent in 2D space in ways that are partially invariant to prompting, with units that encode specific spatial features abstractly.
- We find specific units that predict action correctness during spatial navigation, which are also activated in unrelated spatial reasoning contexts.

2 Related Work

World Models in LLMs. World models have been defined as a compact, coherent, and interpretable representation of the generative process underlying the training data [9, 16, 17]. Some authors argue that LLMs lack internal world models capable of predicting world states and simulating action outcomes, which can impair their performance in agentic and planning tasks [4, 5]. On the other hand, some studies have identified specific neurons that encode space and time, demonstrating internal models that remain robust to variations in prompting [9]. Other investigations have found evidence for internal models of the non-linguistic world—ranging from perceptual structures such as color to spatial orientation concepts, cardinal directions, and object properties [6,8,18]. Implicit world models have also been described in goal-oriented contexts where the representation is influenced by an agent's objectives [19].

Internal Representations and Mechanistic Interpretability. Mechanistic interpretability has emerged as a promising avenue for understanding the inner workings of LLMs [20–22]. This approach has been used to analyze emergent behaviors in larger models and to trace how internal representations influence output, providing essential insights for causality and AI safety [23–25]. For example, research has shown that neurons across different layers tend to specialize, with middle layers often containing neurons that represent higher-level contextual features [26–29]. Furthermore, some neurons exhibit task-specific activations and can be predictive of model performance on these tasks [28, 30–32].

Prompting Techniques and Prompt Influence on Outcome. A growing body of work has established that LLM performance is highly sensitive to the specific prompting techniques employed, across a variety of benchmarks and tasks [12–14]. Even variations in prompt formatting have been found to lead to notable differences in model behavior [33]. This has driven the development of a wide range of prompting techniques which become highly relevant when attempting to improve LLM performance [34, 35].

LLMs for Spatial Navigation. LLMs have also been applied to spatial navigation tasks despite being trained solely on text [36, 37]. These tasks have commonly been represented as text-based sequences, where models are asked to provide information about the environment or actions that have effects in the world [38–40]. Previous studies have evaluated LLMs as agents navigating grid-world environments, where models must plan a full path in advance to locate goals and avoid obstacles [41, 42]. These approaches demonstrate that LLMs can solve spatial navigation tasks, although small models have shown limited generalization to variations such as differing grid sizes or obstacle configurations [41].